

Glucose Monitor Prototype — Project Proposal

Sep 24, 2026 · Prepared by @Kim

Project Overview

This proposal covers the design and build of a **glucose monitoring prototype** built on the ESP32-C3 microcontroller. The device reads two electrochemical sensor channels — **sweat** and **urine** — using glucose oxidase (GOx) enzyme electrodes, and displays live readings on an onboard OLED screen.

This is a **research and educational prototype**, not a certified medical device. It demonstrates the full sensing, signal-processing, and display pipeline in a compact, presentable unit — it is not intended to deliver clinically validated blood glucose readings.

Scope of Work

1. **Firmware development** (ESP32-C3, Arduino/PlatformIO)
 - Dual-channel analog sampling with oversampling for signal stability
 - Two-point calibration framework per channel
 - Live readout on an onboard OLED display
2. **Analog front end** — a discrete op-amp transimpedance amplifier circuit per channel (sweat, urine), keeping the build cost low while avoiding a dedicated potentiostat IC
3. **Circuit assembly** — breadboard prototype, wiring, and bring-up/testing of both sensor channels
4. **Enclosure** — a compact, presentable 3D-printed case with an OLED display window, USB-C access, and electrode wire feedthroughs. **3D modeling of the enclosure will be outsourced to a third-party designer**; this scope covers coordinating that design against the electronics layout and integrating the finished parts.
5. **Documentation** — wiring diagram, bill of materials, and basic operating notes handed over with the finished unit

Deliverables

- ☐ Fully assembled prototype (ESP32-C3 + sweat/urine sensor channels + OLED readout)
- ☐ 3D-printed enclosure, outsourced modeling integrated and fitted to the electronics
- ☐ Firmware source, flashed and working on the delivered unit
- ☐ Wiring diagram
- ☐ Bill of materials
- ☐ Basic operating and calibration notes

Materials & Cost Breakdown

Estimated component cost for one prototype unit, PHP:

Item	Qty
ESP32-C3 development board	1
Dual op-amp IC (LM358/MCP6002)	2
Resistor assortment kit	1
Solderless breadboard	1
Jumper wire set	1
SSD1306 OLED display (0.96", I2C)	1
Glucose test strips (GOx electrode source, box of 50)	1
USB-C / Micro-USB cable	1
Misc. wiring, solder, connectors, contingency	1

Materials quote: PHP 5,000–6,000, covering the items above plus sourcing, handling, and spares.

Timeline

Phase	Activities	Est. Duration
1. Procurement	Order electronics locally, source enzyme electrodes	2–3 weeks
2. Circuit build & firmware	Breadboard assembly, firmware flashing, sensor bring-up, iteration	3–4 weeks
3. Enclosure	Outsourced 3D modeling, revisions, printing, fitting to electronics	3–4 weeks (parallel with Phase 2)
4. Calibration & testing	Two-point calibration, functional testing, refinement	3–4 weeks
5. Handover	Documentation, delivery, walkthrough	1–2 weeks

Estimated total: 3–4 months from project start, accounting for parts availability, iterative calibration, and the outsourced designer's turnaround.

Investment

Total project cost: PHP 18,000, inclusive of materials and labor.

Covers	Notes
Materials	PHP 5,000–6,000, per the breakdown above
Design & firmware development	Circuit design, firmware, calibration framework
Assembly labor	Breadboard build, wiring, sensor bring-up, testing
Enclosure	Outsourced 3D modeling fee + printing, coordinated and integrated by this project
Documentation & handover	Wiring diagram, BOM, operating notes

This is a **flat quote** for one prototype unit as scoped above. Changes to scope (e.g. additional sensor channels, a production-run enclosure, or PCB fabrication in place of the breadboard) would be quoted separately.

Important Scope Note

This device measures a **relative electrochemical signal** from sweat and urine — it is not a substitute for a clinical or consumer blood glucose meter, and results should not be used for medical decisions.

- **Urine glucose** only appears once blood glucose exceeds the renal threshold (~180 mg/dL); below that, it reads near zero. It functions as a threshold indicator, not a continuous measurement.
- **Sweat glucose** concentration is roughly 100x lower than blood glucose and is sensitive to sweat rate and electrode condition, so readings should be treated as a trend signal rather than an absolute number without extensive calibration.

This scope note should be communicated to any end users of the prototype.

Terms & Next Steps

- **Total:** PHP 18,000, materials and labor included
- **Payment terms:** to be confirmed (e.g. 50% down payment to begin procurement, balance on delivery)
- **Scope changes:** any additions beyond this proposal (extra channels, production enclosure, PCB fabrication) will be quoted separately before work begins

To proceed, please confirm acceptance of this proposal and the down payment terms so procurement can begin.